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Estimation of the harvest index and the relative water content – Two examples of composite variables in agronomy

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ABSTRACT

Composite variables are variables derived from measurable traits. They are commonly used in agronomy: two well-known examples being the harvest index and the relative water content. There are two approaches for finding estimated averages of such variables that are derived from direct measurements: They can be found either based on a calculation using individual measurements ("pre-processing") or from a calculation using averages or estimates ("after-fitting"). The former needs to be done prior to fitting a statistical model whereas the latter is carried out after a statistical model has been fitted to the original measurements. We show that the commonly used pre-processing approach results in biased estimates. Moreover, the bias depends on both the correlation between and the uncertainty associated with the variables used for the composite variable. This finding is shown in two examples and a simulation study.

1. Introduction

Recently there has been an increased awareness of the use of appropriate statistical methods in agronomy (Acutis et al., 2012; Jensen et al., 2018; Kitsche and Schaarschmidt, 2015; Ritz et al., 2013; Stroup, 2015). However, so far little attention has been paid to the analysis of composite variables such as ratios, despite the routine use of ratios to estimate a range of crop-specific indices. In other fields, there has been some focus on how to properly deal with composite variables (e.g., Curran-Everett, 2013; Curran-Everett and Williams, 2015; Vickers, 2001).

A well-known and commonly used composite variable in agronomy is the harvest index (HI). The HI was initially defined for cereals as the ratio of grain yield to the total aboveground biomass (Donald and Hamblin, 1976). Later, it has been adapted to other crop species as well (Hay, 1995). The HI is frequently used (Dai et al., 2016; Reynolds et al., 2017) and integrated as an important variable for evaluation of yield gains in breeding programs (Pask et al., 2012; Hütsch and Schubert, 2017).

Another commonly used composite variable in agronomy is the relative water content (RWC), describing plant water status as a ratio that involves fresh weight, dry weight and turgid weight of sampled leaf tissue (Smart and Bingham, 1974). The RWC indicates the water deficit in plant leaves compared to a full turgid pressure and therefore the variable has been commonly used to compare plants with different degree of water-stress in studies related to drought-resilient crops, e.g., in breeding (Hu and Xiong, 2014).

The HI and the RWC are not directly measurable traits of the plant. They are derived from measured morphological or physiological characteristics. Analysis of composite variables, including HI and RWC, may be carried out in two ways: The composite variable may be obtained using calculations carried out before or after fitting a relevant statistical model. However, nonlinear transformations of two or more variables (such as ratios) will entail a loss of information if the statistical analysis is based on measurements that were pre-processed or transformed beforehand compared to combining estimates from model fits based on the original measurements (Kisbu-Sakarya et al., 2014). Combining variables before statistical estimation implies losing some information as the involved variables are 1) subject to measurement error and 2) they may be correlated. These two features will often not be adequately captured when the mathematical manipulations used for combining values precede the statistical analysis and, hence, are not accounted for statistically (Hinkley, 1969; Kisbu-Sakarya et al., 2014). For linear transformations (assuming normally distributed measurements), there is no loss in information.

Admittedly, the pre-processing approach is very convenient, as it only requires simple mathematical manipulations (addition, subtraction, multiplication, and division) of the original measurements, which

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are feasible in any spreadsheet programme. In contrast, the after-fitting approach requires a means to estimate the uncertainty (e.g., standard error) for the averages or even differences of averages of a composite variable: The standard error cannot simply be transformed in the same way as the individual measurements or the averages (van der Vaart, 1998). The statistical methodology needed is referred to as the delta method and it has to be applied to estimates derived from several model fits (one for each variable used in the composite variable). Recent advances in statistical software has made it much easier to apply the delta method in practice without any need to master the underlying mathematical tools (e.g., Jensen and Ritz, 2015).

In this study, we show how to estimate population means for indices and other composite variables without biasing the results. We hypothesize that an analysis based on pre-processed data will result in bias when estimating means of composite variables based on ratios or products of normally distributed variables. To utilize data in an optimal way, a marginal models approach, where a model is fitted to each directly measured variable, is used and the estimated mean for the composite variable is then calculated based on parameter estimates from the resulting model fits. The approach is demonstrated using two examples from the literature. In the first example, we re-evaluate how the HI varies for winter wheat cultivars under different nitrogen regimes. In the second example, we revisit an experiment with quinoa where we estimate and compare the RWC under different drought and fertilizer conditions. Finally, we present results from a simulation study examining the bias and familywise error rate associated with estimating averages of different composite variables.

2. Material and methods

2.1. Data examples

2.1.1. Harvest index and nitrogen content in winter wheat cultivars

We consider data from a field trial investigating four winter wheat cultivars with known differences in rooting depths that were treated with different levels of nitrogen (N) fertilization (Rasmussen et al., 2015). The trial was carried out in the season 2011–2012 and repeated in the season 2012–2013 in conventional agricultural fields at the University of Copenhagen, Taastrup, Denmark (55°40′N; 12°18′E). For each season, the experiment was laid out as a randomized complete block design with four replicates. One aim of this study was to evaluate root growth measured by using minirhizotron tubes (Rasmussen et al., 2015). As it was expected that soil conditions could render underground sampling and/or imaging impossible in some cases, the authors of the original paper ended up aiming for a balanced data set with three replicates. An extra block was included in case of problems with the minirhizotron tubes. In such cases, data from the corresponding cultivar and N combination from the fourth block were used instead to be able to have three replicates for all treatment combinations.

In this study, we only consider harvest data from the season 2011–2012. The different winter wheat cultivars grown were Hereford, classified as fodder wheat; and three bread wheat cultivars, Cordiale, Genius, and JB Asano. An inorganic N fertilizer (50% NO_3^-, 50% NH_4^+) was applied at different levels such that Hereford received 20, 85, 150, 250 or 350 kg N ha^{-1} whereas the three bread wheat cultivars received 85 or 250 kg N ha^{-1}. Levels above 150 kg ha^{-1} were split in two applications of 150 and 100 or 200 kg N ha^{-1}. Plant material was sampled in August at harvest (BBCH 85–88). Plant material was separated in grain and straw yield, dried at 80 °C for 24 h, weighted, milled and analyzed for percentage N content using an Organic Elemental analyzer, Flash 2000. We focused on the HI defined as a ratio by the following equation:

$$\mathrm{HI} = \frac{\mathrm{grain\ DW}}{\mathrm{grain\ DW} + \mathrm{straw\ DW}},$$

with dry weight (DW) grain and straw reported in kg ha^{-1}, and the N content in grain defined as a product by the following equation:

$$\mathrm{N} = \mathrm{N\%} \bullet \mathrm{grain\ DW}$$

where N% is the measured percentage N in the grain yield and DW grain yield is reported in kg ha^{-1}. Thus, both the HI and N content are composite variables derived from two measured variables.

2.1.2. Relative water content for quinoa in different nitrogen and water regimes

We considered data from a study on the effect of N fertilization and drought in quinoa (Alandia et al., 2016). The experiment was carried out from June to August in a greenhouse at the University of Copenhagen, Taastrup, Denmark (55°40′N; 12°18′E). Day and night temperatures were 24 ± 2 °C and 21 ± 2 °C, respectively, with a relative humidity of 20–86% (56.2% ± 14). The total incoming radiation came from natural sunlight with a photoperiod of 17.5 – 15 h (June - August). Three seeds of quinoa (*Chenopodium quinoa* Willd.) cv. Titicaca were sown in each pot (25 cm tall by 16 cm diameter) and thinned to one plant at the two-leaf stage. Local sub-soil classified as coarse sand was used in the pots. The soil surface was covered with 4 cm (1 kg) of quartz gravel to minimize soil evaporation. All pots received a nutrient solution containing 5.4 g mono-potassium phosphate, 12.5 g magnesium sulphate, 21.8 g potassium chloride, and 22.7 ml pioneer micro + Fe mixed in 1 L of water to meet nutrient requirements for plant growth. Following the base nutrient, one of four N levels was applied: 0, 0.2, 0.4 or 0.6 g N per pot supplied as ammonium nitrate (NH_4NO_3). Before the establishment of water regimes, all pots were fully irrigated to obtain full water holding capacity. Forty days after sowing, during floral bud formation at developmental stages 3–4 (Jacobsen and Stølen, 1993), plants were exposed to one of two water regimes. Either full irrigation, where the plants were irrigated daily at 4 pm to 98% water holding capacity throughout the experiment; or progressive drought, where irrigation was withheld until the transpiration rate of the plant reached 10% of the control, after which the plants were re-watered to 98% water holding capacity for seven days to study the recovery. Four replicates of each treatment combination were used for the experiment and a randomized complete design was aimed at by re-arranging pots daily. Bordering plants surrounded the experiment.

We looked at the relative water content (RWC) defined by this equation:

$$\mathrm{RWC} = \frac{\mathrm{FW} - \mathrm{DW}}{\mathrm{TW} - \mathrm{DW}}$$

where FW was the fresh weight of the leaf in g, TW was the turgid weight in g measured after floating the leaf 4 h under dim light, and DW was the dry weight in g found by drying the leaf for 24 h at 80 °C until constant weight. FW, DW, and TW were measured on half a leaf at two different time points: At the end of the progressive drought and after seven days of recovery from re-watering. Thus, the RWC is a composite variable derived from three measured variables.

2.2. Estimation of composite variables

We consider the scenario where two variables are directly observed in a field or greenhouse experiment, but the interest lies in a composite variable obtained using the nonlinear transformation f of the two variables; the approach to be described later can be extended to more than two variables.

Estimation of mean values for composite variables can be performed using one of two approaches; either in a pre-processing or an after-fitting step. The difference between the two estimates essentially comes down to either estimating the transformed means or estimating the mean of transformed variables.

Estimation of the transformed means has to be carried out using joint or marginal models. The latter utilizes estimates from several marginal models (one per variable used in the composite variable) and in this study we will refer to it as the marginal models approach. The former requires a simultaneous (multivariate) model for all variables. In this study, we only consider the marginal models approach as it provides higher flexibility and has less computational problems in terms of convergence (Jensen and Ritz, 2018).

Estimation of the mean of transformed variables, referred to as the pre-processing approach, is carried out fitting a statistical model to data obtained by applying the transformation to each individual measurement.

In a more mathematical formulation, the two situations correspond to estimating $f(E(X,Y))$ or $E(f(X,Y))$ where E refers to the mean. Unless f is a linear function, the two estimates do not agree. Also, in case f is a nonlinear function and the pair (X,Y) follows a bivariate normal distribution (which may be the case quite often) then the transformation $f(X,Y)$ will not follow a normal distribution. However, the pre-processing approach would usually imply that this transformation is normally distributed. Consequently, the statistical model usually assumed in the pre-processing approach is not correctly specified.

2.2.1. The pre-processing approach

As already mentioned, the pre-processing approach for estimating mean levels of a composite variable implies calculating the value of the variable for each individual measurement before doing any statistical analysis. Using HI as an example, this corresponds to making a new variable by dividing measured grain yield by measured total biomass yield for each experimental unit, i.e. for each plot. The estimated mean HI level for each treatment combination is then obtained using a standard analysis of variance (ANOVA) type of model. Specifically, for a factorial block design with replicates (e.g., Example 1, section 3.1) we consider the following linear mixed ANOVA model (Oehlert, 2010):

$$\mathrm{HI}_{ijk} = \frac{\text{grain DW}_{ijk}}{\text{straw DW}_{ijk} + \text{grain DW}_{ijk}} = (\text{Cultivar, N})_j + Block_k + \varepsilon_{ijk}, \tag{1}$$

where HI_{ijk} is the HI for plot i for cultivar-N combination j in block k. The variable HI_{ijk}, is a function of the two measurable variables, grain DW_{ijk} and straw DW_{ijk}, respectively. HI_{ijk} is, therefore, not directly measurable. The parameter $(\text{Cultivar, N})_j$ is the effect of the cultivar-N combination at level j, $Block_k$ is the random block effect for block k, and ε_{ijk} is the (propagated) residual error of HI_{ijk}. The ε_{ijk}' s and B_k' s are assumed to be mutually independent normally distributed variables, $\varepsilon_{ijk} \sim N(0, \sigma^2)$ and $B_k \sim N(0, \sigma^2_{block})$ with unknown variance parameters, σ^2 and σ^2_{block}, respectively. Similarly, the model for the composite variable N content in grain (also considered in Example 1, section 3.1) is formulated as:

$$\text{N grain}_{ijk} = \text{N\% grain}_{ijk} * \text{grain DW}_{ijk} = (\text{Cultivar, N})_j + Block_k + \varepsilon_{ijk}, \tag{2}$$

The model for RWC (considered in Example 2, section 3.2) is formulated as:

$$\mathrm{RWC}_{ij} = \frac{\mathrm{FW}_{ij} - \mathrm{DW}_{ij}}{\mathrm{TW}_{ij} - \mathrm{DW}_{ij}} = (\text{N, Water})_j + \varepsilon_{ij}, \tag{3}$$

where RWC_{ij} is the RWC for leaf i for N-water treatment combination j, and, $\varepsilon_{ij} \sim N(0, \sigma^2)$.

The very common assumption of normally distributed measurements is actually a misspecification of the model if the measured variables are assumed to follow a multivariate normal distribution. However, the normal assumption may be difficult to reject based on residual and QQ-plots and it may still be an acceptable approximation. Estimated mean levels for specific treatment combinations are then obtained directly from the resulting linear mixed model fit.

2.2.2. The after-fitting approach

Alternatively, estimated mean levels of composite variables may be obtained using an after-fitting approach where a (marginal) model is fitted for each observed variable. Subsequently, estimated mean levels of the composite variable are found by combining estimated mean levels of the variables involved (in an after-fitting step). More specifically, for our example on HI, we fit separate marginal models for the two variables, straw and grain yield. The estimated means for each factor level of the grain yield and the straw yield, respectively, are obtained directly from the corresponding models. Assuming a factorial block design, the corresponding linear mixed model for each of the directly measured variables can be written in the following way

$$\text{grain DW}_{ijk} = (\text{Cultivar, N})_j + Block_k + \varepsilon_{ijk}, \tag{2}$$

$$\text{straw DW}_{,ijk} = (\text{Cultivar, N})_j + Block_k + \varepsilon_{ijk}, \tag{3}$$

similarly to Eq. (1), with the same assumptions in place for each of the two models.

Finally, estimated mean levels of the composite variable are obtained by combining estimates from the two models, (2) and (3), in terms of a function, f, of the involved parameters:

$$\mathrm{HI}_{(cultivar,N)j} = f(\text{grain DW}_{(cultivar,N)j}, \text{straw DW}_{(cultivar,N)j}) = \frac{\text{grain DW}_{(cultivar,N)j}}{\text{straw DW}_{(cultivar,N)j} + \text{grain DW}_{(cultivar,N)j}}$$

Standard errors for these estimates or differences of such estimates are obtained using the delta method. The delta method is based on a Taylor series expansion and is an asymptotic approximation, which may lead to slightly biased standard errors in case of small sample sizes. In the case of independent variables, the delta method only requires the estimated means and the corresponding standard errors as arguments. In case of dependent variables, the delta method needs the joint estimated variance-covariance matrix for all parameters involved in the derived variable. In case all parameters are obtained from the same model fit, the covariance matrix is estimated as part of the regular model fitting procedure. However, for the more general setting where the parameters are obtained from several marginal models, an extended version of the delta method must be used (Jensen and Ritz, 2015).

2.3. Statistical analysis

2.3.1. Harvest index and nitrogen content in winter wheat cultivars

For each of the three variables, grain DW, straw DW and percentage N content in grain, data were analyzed using a linear mixed one-way ANOVA model of the form (2) with the combination of cultivar and N level as fixed effects and block as random effect. The two composite variables, HI and N content in grain, were estimated for each factor level using the pre-processing approach and the after-fitting approach as described in Section 2.2.1 and 2.2.2. Results were reported as estimates with 95% confidence intervals. Model assumptions were assessed by visual inspection of residuals and predicted random effects in residual- and QQ-plots. N percentage and N content in grain were log-transformed before analysis. For these variables, estimates and 95% confidence intervals were obtained on the transformed scales and back-transformed subsequently.

2.3.2. Relative water content for quinoa in different nitrogen and water regimes

For each sampling time, data on FW, DW, and TW were analyzed using a one-way ANOVA with combinations of N and water treatment as the factor levels. RWC was estimated for each factor level using both the pre-processing approach and the after-fitting approach as described in Section 2.2.1 and 2.2.2. Results were reported as estimates with 95% confidence intervals. Model assumptions were assessed by visual inspection of residuals in residual- and QQ plots. For the second time point, i.e., after seven days of recovery, FW, DW, and TW were log-transformed before analysis to obtain variance homogeneity. For these variables, estimates and 95% confidence intervals were obtained on the

Table 1
Estimated harvest index (HI) and the total N content (kg ha^{-1}) in the grain from the winter wheat experiment with 95% confidence intervals. HI and total N from grain were estimated by making a new variable before analysis (pre-processing) and by combining estimates from marginal model fits (after-fitting). Within each variable and analysis approach, treatments with the same super-script letter were not significantly different.

Cultivar	App. N (kg ha^{-1})	N grain (pre-processing) (kg ha^{-1})		N grain (after-fitting) (kg ha^{-1})		HI (pre-processing) (%)		HI (after-fitting) (%)	
Cordiale	85	82.0	$(60.5\text{-}103.5)^{bc}$	86.7	$(65.8\text{-}107.7)^{b}$	52.0	$(50.6\text{-}53.4)^{bc}$	53.6	$(50.8\text{-}56.4)^{b}$
	250	208.6	$(187.1\text{-}230.1)^{d}$	217.2	$(193.7\text{-}240.7)^{e}$	56.3	$(54.9\text{-}57.7)^{f}$	57.2	$(55.1\text{-}59.3)^{dg}$
Genius	85	76.9	$(55.4\text{-}98.4)^{bc}$	82.5	$(69.1\text{-}95.9)^{b}$	50.0	$(48.6\text{-}51.3)^{a}$	51.7	$(48.6\text{-}54.8)^{a}$
	250	194.7	$(173.2\text{-}216.2)^{d}$	202.5	$(176.0\text{-}229.1)^{de}$	54.5	$(53.1\text{-}55.8)^{de}$	55.4	$(53.6\text{-}57.2)^{cdf}$
Hereford	20	46.3	$(24.8\text{-}67.8)^{a}$	51.2	$(41.6\text{-}60.8)^{a}$	53.3	$(52.0\text{-}54.7)^{cd}$	55.7	$(52.0\text{-}59.3)^{cde}$
	85	75.5	$(54.0\text{-}97.0)^{ab}$	79.8	$(69.9\text{-}89.7)^{b}$	53.1	$(51.7\text{-}54.5)^{cd}$	54.4	$(52.0\text{-}56.7)^{bc}$
	150	106.8	$(85.3\text{-}128.3)^{c}$	111.6	$(98.9\text{-}124.3)^{c}$	55.1	$(53.7\text{-}56.5)^{ef}$	56.0	$(54.2\text{-}57.8)^{eg}$
	250	189.4	$(167.9\text{-}210.9)^{d}$	196.5	$(176.0\text{-}217.0)^{d}$	56.6	$(55.2\text{-}57.9)^{f}$	57.3	$(55.1\text{-}59.4)^{eg}$
	350	208.8	$(187.3\text{-}230.3)^{d}$	216.8	$(193.5\text{-}240.1)^{de}$	55.2	$(53.8\text{-}56.5)^{ef}$	56.0	$(54.4\text{-}57.6)^{ef}$
JB Asano	85	80.3	$(58.7\text{-}101.8)^{bc}$	85.2	$(71.0\text{-}99.4)^{b}$	50.0	$(48.7\text{-}51.4)^{a}$	51.5	$(48.8\text{-}54.1)^{a}$
	250	189.4	$(167.9\text{-}210.9)^{d}$	186.3	$(154.5\text{-}218.0)^{d}$	51.2	$(49.8\text{-}52.5)^{ab}$	51.1	$(49.6\text{-}52.6)^{a}$

Table 2
Estimated relative water content (RWC) with 95% confidence intervals at the end of progressive drought treatment of quinoa plants exposed to four nitrogen concentrations and two hydric levels: Drought (D) and Water (W). RWC levels were estimated using the pre-processing approach and the after-fitting approach. Within each analysis approach, treatments with the same super-script letter were not significantly different.

Water	N (g $pot^{-1)}$	RWC (pre-processing)		RWC (after-fitting)	
W	0	0.892	$(0.827\text{-}0.957)^{cd}$	0.886	$(0.843\text{-}0.929)^{d}$
	0.2	0.931	$(0.865\text{-}0.996)^{d}$	0.929	$(0.870\text{-}0.988)^{d}$
	0.4	0.881	$(0.815\text{-}0.946)^{cd}$	0.882	$(0.870\text{-}0.894)^{d}$
	0.6	0.814	$(0.749\text{-}0.879)^{bc}$	0.814	$(0.800\text{-}0.829)^{c}$
D	0	0.815	$(0.750\text{-}0.881)^{bc}$	0.828	$(0.741\text{-}0.914)^{bcd}$
	0.2	0.736	$(0.670\text{-}0.801)^{ab}$	0.731	$(0.659\text{-}0.803)^{b}$
	0.4	0.779	$(0.714\text{-}0.845)^{b}$	0.776	$(0.736\text{-}0.816)^{bc}$
	0.6	0.643	$(0.578\text{-}0.708)^{a}$	0.640	$(0.588\text{-}0.692)^{a}$

Table 3
Estimated relative water content (RWC) with 95% confidence intervals after seven days of recovery of quinoa plants exposed to four nitrogen concentrations and two hydric levels: Drought (D) and Water (W). RWC levels were estimated using the pre-processing approach and the after-fitting approach. Within each analysis approach, treatments with the same super-script letter were not significantly different.

Water	N (g $pot^{-1)}$	RWC (pre-processing)		RWC (after-fitting)	
W	0	0.880	$(0.857\text{-}0.904)^{ab}$	0.880	$(0.874\text{-}0.887)^{ab}$
	0.2	0.889	$(0.866\text{-}0.913)^{ab}$	0.889	$(0.881\text{-}0.898)^{bc}$
	0.4	0.895	$(0.872\text{-}0.919)^{ac}$	0.895	$(0.882\text{-}0.908)^{c}$
	0.6	0.864	$(0.840\text{-}0.887)^{a}$	0.863	$(0.845\text{-}0.882)^{a}$
D	0	0.889	$(0.866\text{-}0.913)^{ab}$	0.889	$(0.876\text{-}0.902)^{bc}$
	0.2	0.896	$(0.873\text{-}0.920)^{ac}$	0.896	$(0.874\text{-}0.918)^{bc}$
	0.4	0.905	$(0.881\text{-}0.929)^{bc}$	0.905	$(0.890\text{-}0.919)^{c}$
	0.6	0.925	$(0.901\text{-}0.948)^{c}$	0.924	$(0.882\text{-}0.966)^{c}$

transformed scale and back-transformed.

All analyses were carried out in the statistical programming software R version 3.4.2 (R Core Team, 2017). In particular, the extension packages *lme4* (Bates et al., 2015) and *multcomp* (Hothorn et al., 2008) were used for fitting linear mixed models and obtaining pairwise comparisons, respectively.

3. Results

3.1. Harvest index and nitrogen content in winter wheat cultivars

For each cultivar, the DW grain and straw yield increased with the level of applied N except for the highest level of applied N for the cultivar Hereford. Similarly, the N percentage content in grain increased with higher levels of applied N. For a more detailed discussion of the results, we refer to Rasmussen et al., 2015. Estimated DW grain yield, DW straw yield, and percentage N in grain for each cultivar and fertilizer combination can be found in Table S1 in the supplementary material.

Table 1 shows HI and N content in grain estimated using the pre-processing and the after-fitting approach for each cultivar and fertilizer combination. For both HI and N content in grain, the estimated mean levels differed between the two approaches. Overall, the after-fitting approach resulted in larger estimates than the pre-processing approach. For large estimates, the corresponding confidence intervals were in general wider for the pre-processing approach as compared to those from the after-fitting approach, whereas the opposite was true for confidence intervals associated with the smaller estimates. The difference between the two approaches was also evident in the pairwise comparisons of treatments where conclusions depended on the approach (Table S2 in the supplementary material). For example, for the pre-processing approach, a significantly higher HI was observed for Cordiale compared to Genius when applying 250 kg ha^{-1} N. A similar difference was not observed for the after-fitting approach. Most of the pairwise comparisons resulted in quite similar estimated differences for the two approaches. However, for some comparisons the difference was noticeable. In addition, the confidence intervals for the estimated differences were in general wider for the pre-processing approach as compared to the after-fitting approach.

3.2. Relative water content for quinoa in different nitrogen and water regimes

Tables 2 and 3 present the estimated RWC at the end of the progressive drought and after seven days of recovery, respectively. Estimates of RWC were very similar between analysis strategies. Estimates of RWC after recovery differed between analysis strategies but for most of the treatments only on the fourth decimal (not shown). For most treatment combinations, confidence intervals were smaller for the after-fitting approach. Accordingly, conclusions regarding comparisons of the RWC between treatments also depended on the analysis strategy: at the end of progressive drought treatment, the pre-processing analysis strategy showed a significant difference in RWC in the well-watered plants between N-treatments of 0.2 and 0.6 g per pot. The after-fitting approach, on the other hand, showed significant differences between the 0.6 g per pot N-level treatment and all other N-level treatments of well-watered plants. After seven days of recovery, the pre-processing approach showed no differences between N-levels for the well-watered plants, while the after-fitting approach showed significant differences between the well-watered plants with N-level 0.6 g per pot and the two medium N-level treatments of 0.2 and 0.4 g per pot (Table 3 and Table S3). Overall, the estimates from the two modeling approaches were close, but the confidence intervals were markedly different. For more discussion on the biological interpretation of the results, we refer to

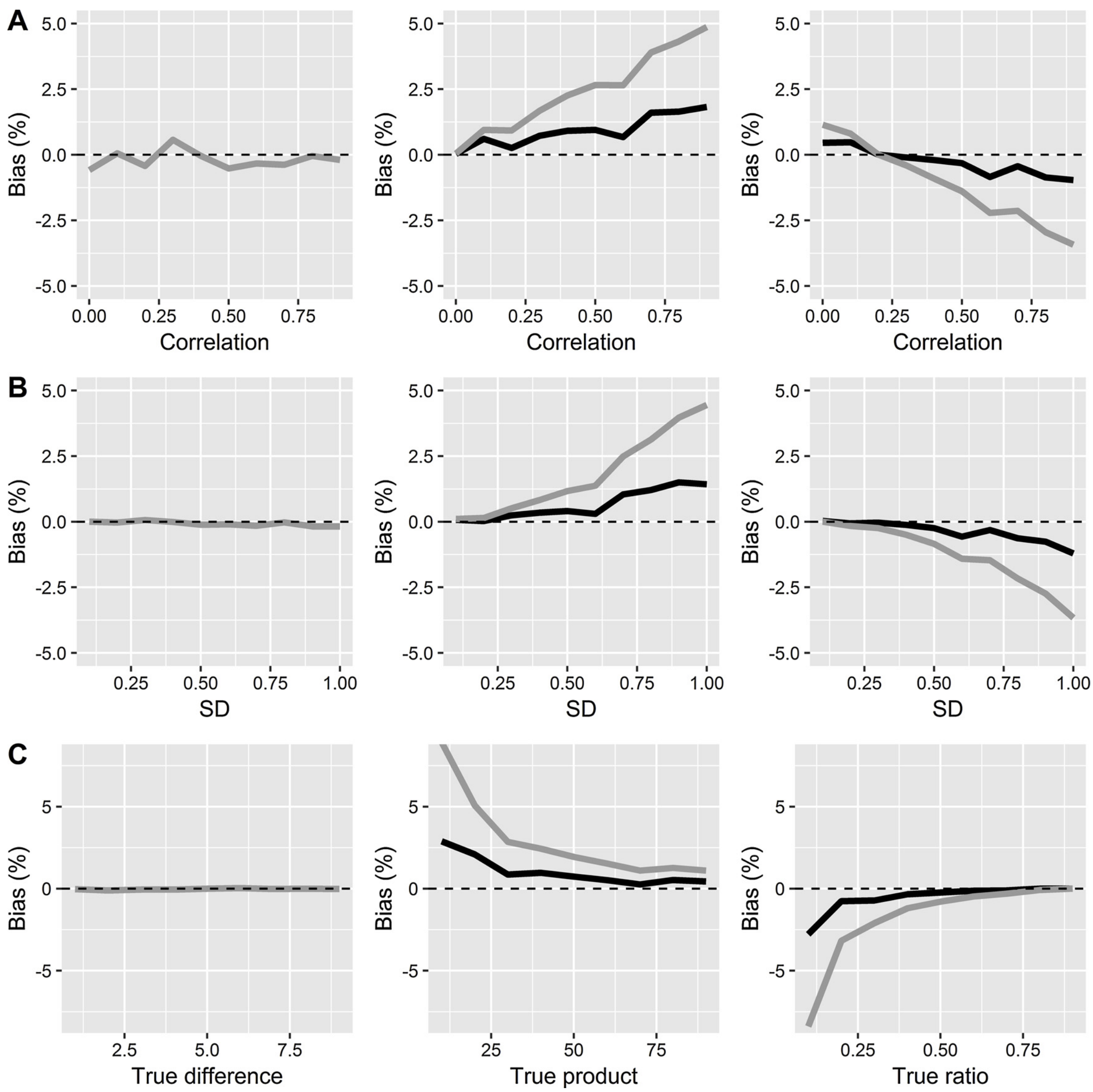

Fig. 1. Percentage bias from the simulation study with N = 3 replicates. Left column: Estimated difference (a linear transformation). Middle column: Estimated product. Right column: Estimated ratio. The grey line represents results from the pre-processing approach. The black line represents results from the after-fitting approach. Panel A, B, and C correspond to increasing correlations, increasing (common) standard error for the two variables, and increasing value of the parameter of interest, respectively.

Alandia et al., 2016.

3.3. Simulation study

To substantiate the results from the two examples, a simulation study was carried out. The simulations were based on two normally distributed variables, Y_1 and Y_2. As the starting point for the simulations, we considered the scenario where Y_1 and Y_2 had mean levels 2 and 10, respectively, with a common standard deviation equal to 1, a correlation between Y_1 and Y_2 equal to 0.9 and with either 3 or 10 replicates. Data were analyzed using the pre-processing and the after-fitting approach in a scenario with an ANOVA model with a single group, corresponding to estimating a single mean and standard error. For increasing common standard deviation (0.1, 0.5, 1, 3, 5), increasing correlations between Y_1 and Y_2 (0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1), and an increasing mean level (1,…,10) of the first variable, Y_1, we estimated the difference, the product, and the ratio. The true mean values for the different composite variables were calculated based on the known, true means of the two measured variables, $E(Y_1)$ and $E(Y_2)$. Ten thousand data sets were generated for each scenario and results were reported graphically as percentage bias.

For a subset of all possible scenarios presented above, where mean value of $Y_1 = 2$ or 5, mean value of $Y_2 = 10$, common standard deviation = 0.1, 1, or 3, correlation between Y_1 and $Y_2 = 0$, 0.3, or 0.9, and number of replicates N = 5, 20, 100, results were presented as percentage bias, standard error and coverage. Coverage was defined as the percentage of times the true value was within the confidence interval; consequently, for 95% confidence intervals, the coverage should be close to 0.95.

Fig. 1 and 2 show percentage bias for 3 and 10 replicates, respectively. Percentage bias, standard errors, and coverage for different combinations of the parameters with 5, 20 and 100 replicates can be found in Tables S4 and S5 in the supplementary material.

No bias was observed for any of the two approaches when considering the difference, a linear function of parameters (Fig. 1 and 2, left column). However, for the two nonlinear functions, the results differed between approaches: Overall, the pre-processing approach suffered from serious bias in most scenarios, whereas the after-fitting approach consistently resulted in estimates closer to the true value. For the pre-processing approach, the smallest bias was observed for both

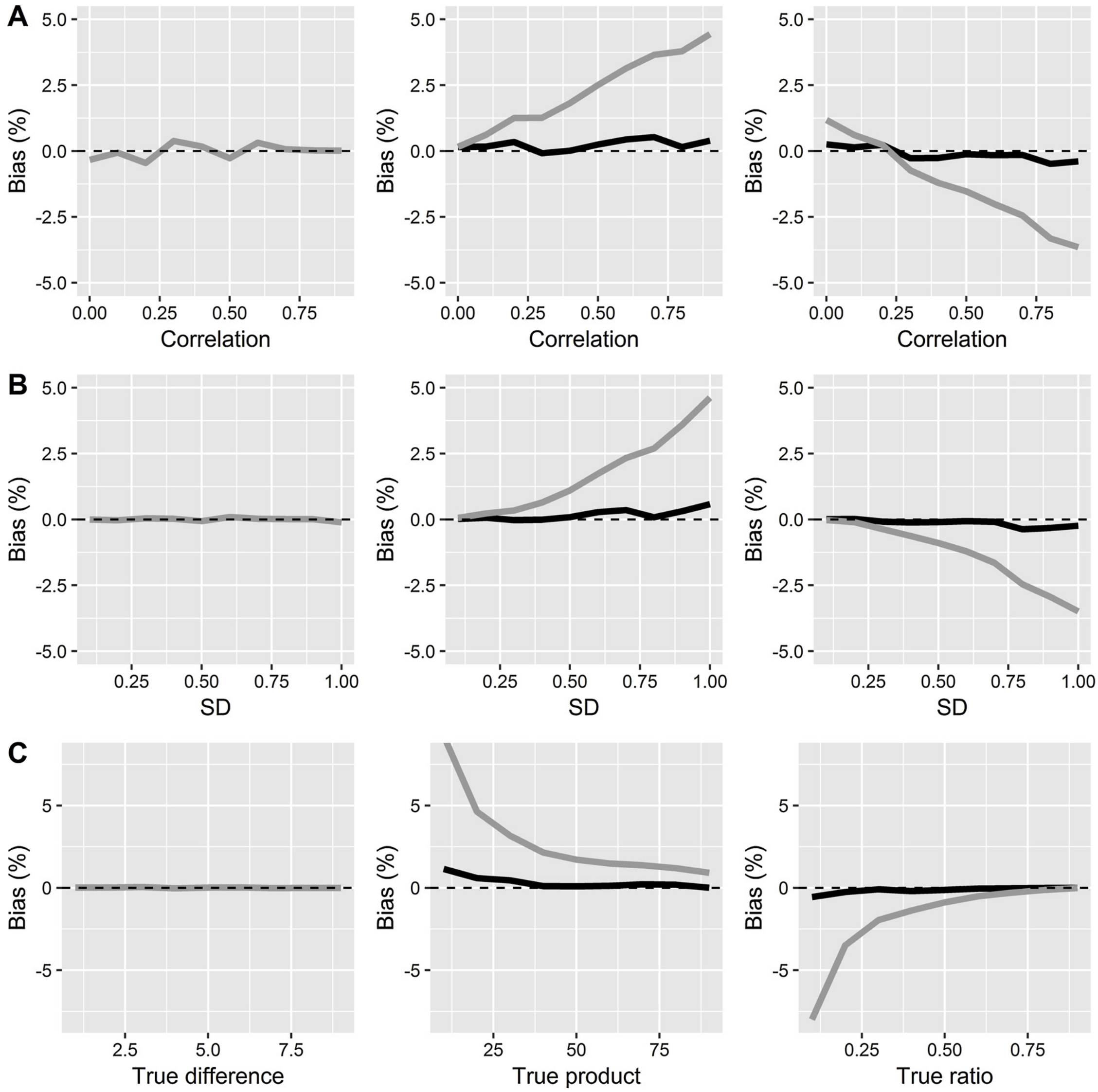

Fig. 2. Percentage bias from the simulation study with N = 10 replicates. Left column: Estimated difference (a linear transformation). Middle column: Estimated product. Right column: Estimated ratio. The grey line represents results from the pre-processing approach. The black line represents results from the after-fitting approach. Panel A, B, and C correspond to increasing correlations, increasing (common) standard error for the two variables, and increasing values of the true parameter, respectively.

the product and the ratio when the measured variables, Y_1 and Y_2, were uncorrelated and observed with high precision, i.e., a small standard deviation. Also for the pre-processing approach, a decreasing percentage bias was observed for increasing mean level of the variable Y_1 (Fig. 1, C, and Fig. 2, C). For both approaches, a certain number of replicates were required to stabilize the results. For increasing sample size, bias vanished for the after-fitting approach while persisting for the pre-processing approach.

For small sample sizes, the estimated standard errors were always smaller for the after-fitting approach as compared to the pre-processing approach (Table S4 and S5). For both approaches, the coverage was below the nominal level of 95% for small sample sizes. For large sample sizes, the after-fitting approach reached the nominal level in all scenarios. The pre-processing approach was far below the nominal level for scenarios where the variables were highly correlated and/or measured with large uncertainty.

Overall, we found similar results when introducing random block effects (results not shown).

4. Discussion

Previous studies have shown that analysis of composite variables involving a change, e.g., percentage change over time or to a reference treatment, is statistically inefficient when using the pre-processing approach (Vickers, 2001). Our study extends these findings to composite variables commonly used in agronomy. Combining observations from several measurements into an index or another composite variable before analysis implies a loss of information.

The composite variables HI and RWC considered in this study are good examples of composite variables used in agronomy. Other composite variables that are derived by combining different directly measurable traits include tolerance to weed control (Rasmussen et al., 2008, 2004), the fraction of transpired soil water (Alandia et al., 2016; Liu and Stützel, 2002), the land equivalent ratio in intercropping (Andersen et al., 2014; Hamzei and Seyedi, 2015), and nitrogen use efficiency and its components (Bingham et al., 2012). We expect that our results also are applicable to other composite variables being nonlinear functions of measurable variables.

Estimating the uncertainty of composite variables have been the focus of earlier published work (Dilba et al., 2007; Geary, 1930; Hinkley, 1969; Kisbu-Sakarya et al., 2014; Schaarschmidt and Djira, 2016). We chose to base our uncertainty estimates on a simple asymptotic approach that indeed showed very good performance in all the examined simulation scenarios, indicating its appropriate use for large sample data sets. As an alternative to the delta method chosen here, different bootstrap strategies may be applied. Such a strategy has the advantage of giving not only an estimate of the index but also of the distribution of the index, and may even be used to validate the deviation from the alternative assumed distribution (Davison and Hinkley, 2007). However, for small datasets, the bootstrap exhibits the same shortcomings as the delta method we used.

It could be argued that most crop specific indices are prone to error or uncertainty from the measurements. For HI this could be a loss of grain during harvest or samples contaminated with soil and other residues that affect straw weight. Also, there may be a large variation in the partitioning of assimilates to below-ground biomass (Hay, 1995) which would potentially bias the measurements. Thus, a little more uncertainty/bias due to the statistical method used may therefore be inconsequential. However, most of the uncertainty related to the measurements have become easier to control with the current advanced technology and, specifically, in a breeding process where the main focus is to compare varieties internally. As the measurements become more precise, the correct statistical analysis becomes more important. For composite variables, this means using the after-fitting approach.

It should be noted that the results rely on assuming normality distributions for the measured variables. Such assumptions may be justified by assuming that the measurement apparatus and device used in the field or greenhouse or elsewhere (including how scientists handle it) outputs quantities that are approximately normally distributed even though they may themselves have been computed using other underlying variables. This approximation seems sensible in the view of methods currently used in agronomy.

Finally, specifically for the ratio of two normally distributed variables, it could happen that the denominator becomes close to 0, resulting in very large absolute values of the ratio, which would then follow a skewed distribution instead of a (symmetric) normal distribution. Moreover, the ratio may not even be well defined if the denominator becomes equal to 0. In such cases, the ratio may not be a good choice for a meaningful composite variable. In practice, ratios, however, are often defined using variables that cannot attain a zero value (e.g., restricted to be positive or negative). In that case, there is no problem with the ratio, but then the measured variables strictly speaking cannot follow normal distributions. Therefore, the after-fitting approach would lead to a slightly misspecified statistical model, but it may still be a useful approximation.

5. Conclusion

As composite variables, like HI and RWC, are used frequently in agronomy and breeding for decision making, it is critically important to use statistical methodology that provides as unbiased and precise estimates as possible. The use of the after-fitting approach is one contribution in this direction.

Declaration of Competing Interest

None

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.eja.2019.125962.

References

Acutis, M., Scaglia, B., Confalonieri, R., 2012. Perfunctory analysis of variance in agronomy, and its consequences in experimental results interpretation. Eur. J. Agron. 43, 129–135. https://doi.org/10.1016/j.eja.2012.06.006.

Alandia, G., Jacobsen, S.-E., Kyvsgaard, N.C., Condori, B., Liu, F., 2016. Nitrogen sustains seed yield of quinoa under intermediate drought. J. Agron. Crop Sci. 202, 281–291. https://doi.org/10.1111/jac.12155.

Andersen, S.N., Dresbøll, D.B., Thorup-Kristensen, K., 2014. Root interactions between intercropped legumes and non-legumes—a competition study of red clover and red beet at different nitrogen levels. Plant Soil 378, 59–72. https://doi.org/10.1007/s11104-013-2014-4.

Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-effects models using lme4. J. Stat. Softw. 67, 1–48. https://doi.org/10.18637/jss.v067.i01.

Bingham, I.J., Karley, A.J., White, P.J., Thomas, W.T.B., Russell, J.R., 2012. Analysis of improvements in nitrogen use efficiency associated with 75 years of spring barley breeding. Eur. J. Agron. 42, 49–58. https://doi.org/10.1016/j.eja.2011.10.003.

Curran-Everett, D., 2013. Explorations in statistics: the analysis of ratios and normalized data. Adv. Physiol. Educ. 37, 213–219. https://doi.org/10.1152/advan.00053.2013.

Curran-Everett, D., Williams, C.L., 2015. Explorations in statistics: the analysis of change. Adv. Physiol. Educ. 39, 49–54. https://doi.org/10.1152/advan.00018.2015.

Dai, J., Bean, B., Brown, B., Bruening, W., Edwards, J., Flowers, M., Karow, R., Lee, C., Morgan, G., Ottman, M., Ransom, J., Wiersma, J., 2016. Harvest index and straw yield of five classes of wheat. Biomass Bioenergy 85, 223–227. https://doi.org/10.1016/j.biombioe.2015.12.023.

Davison, A.C., Hinkley, D.V., 2007. Bootstrap Methods and Their Application, 7th ed. Cambridge University Press, New York.

Dilba, G., Schaarschmidt, F., Hothorn, L.A., 2007. Inferences for Ratios of Normal Means.

Donald, C.M., Hamblin, J., 1976. The Biological Yield and Harvest Index of Cereals As Agronomic and Plant Breeding Criteria. pp. 361–405. https://doi.org/10.1016/S0065-2113(08)60559-3.

Geary, R.C., 1930. The frequency distribution of the quotient of two normal variates. J. R. Stat. Soc. 93, 442–446. https://doi.org/10.2307/2342070.

Hamzei, J., Seyedi, M., 2015. Evaluation of the effects of intercropping systems on yield performance, land equivalent ratio, and weed control efficiency. Agric. Res. 4, 202–207. https://doi.org/10.1007/s40003-015-0161-y.

Hay, R.K.M., 1995. Harvest index: a review of its use in plant breeding and crop physiology. Ann. Appl. Biol. 126, 197–216. https://doi.org/10.1111/j.1744-7348.1995.tb05015.x.

Hinkley, D.V., 1969. On the ratio of two correlated normal random variables. Biometrika 56, 635–639. https://doi.org/10.1093/biomet/56.3.635.

Hothorn, T., Bretz, F., Westfall, P., 2008. Simultaneous inference in general parametric models. Biom. J. 50, 346–363. https://doi.org/10.1002/bimj.200810425.

Hu, H., Xiong, L., 2014. Genetic engineering and breeding of drought-resistant crops. Annu. Rev. Plant Biol. 65, 715–741. https://doi.org/10.1146/annurev-arplant-050213-040000.

Hütsch, B.W., Schubert, S., 2017. Harvest index of maize (Zea mays L.): are there possibilities for improvement? Adv. Agron. 146, 37–82. https://doi.org/10.1016/BS.AGRON.2017.07.004.

Jacobsen, S.-E., Stølen, O., 1993. Quinoa - Morphology, phenology and prospects for its production as a new crop in Europe. Eur. J. Agron. 2, 19–29. https://doi.org/10.1016/S1161-0301(14)80148-2.

Jensen, S.M., Ritz, C., 2018. A comparison of approaches for simultaneous inference of fixed effects for multiple outcomes using linear mixed models. Stat. Med. 37, 2474–2486. https://doi.org/10.1002/sim.7666.

Jensen, S.M., Ritz, C., 2015. Simultaneous inference for model averaging of derived parameters. Risk Anal. 35, 68–76. https://doi.org/10.1111/risa.12242.

Jensen, S.M., Schaarschmidt, F., Onofri, A., Ritz, C., 2018. Experimental design matters for statistical analysis: how to handle blocking. Pest Manag. Sci. 74, 523–534. https://doi.org/10.1002/ps.4773.

Kisbu-Sakarya, Y., MacKinnon, D.P., Miočević, M., 2014. The distribution of the product explains normal theory mediation confidence interval estimation. Multivariate Behav. Res. 49, 261–268. https://doi.org/10.1080/00273171.2014.903162.

Kitsche, A., Schaarschmidt, F., 2015. Analysis of statistical interactions in factorial experiments. J. Agron. Crop Sci. 201, 69–79. https://doi.org/10.1111/jac.12076.

Liu, F., Stützel, H., 2002. Leaf water relations of vegetable amaranth (Amaranthus spp.) in response to soil drying. Eur. J. Agron. 16, 137–150. https://doi.org/10.1016/S1161-0301(01)00122-8.

Oehlert, G., 2010. A First Course in Design and Analysis of Experiments. W. H. Freeman New York.

Pask, A., Pietragalla, J., Mullan, D., 2012. Physiological breeding II: a field guide to wheat phenotyping. Chemistry (Easton). https://doi.org/10.1017/CBO9781107415324.004.

R Core Team, 2017. R: A Language and Environment for Statistical Computing.

Rasmussen, I.S., Dresbøll, D.B., Thorup-Kristensen, K., 2015. Winter wheat cultivars and nitrogen (N) fertilization—effects on root growth, N uptake efficiency and N use efficiency. Eur. J. Agron. 68, 38–49. https://doi.org/10.1016/j.eja.2015.04.003.

Rasmussen, J., Bibby, B.M., Schou, A.P., 2008. Investigating the selectivity of weed harrowing with new methods. Weed Res. https://doi.org/10.1111/j.1365-3180.

2008.00661.x.
Rasmussen, J., Kurtzmann, J.I., Jensen, A., 2004. Tolerance of competitive spring barley cultivars to weed harrowing. Weed Res. 44, 446–452. https://doi.org/10.1111/j.1365-3180.2004.00419.x.
Reynolds, M.P., Pask, A.J.D., Hoppitt, W.J.E., Sonder, K., Sukumaran, S., Molero, G., Pierre, C., Saint Payne, T., Singh, R.P., Braun, H.J., Gonzalez, F.G., Terrile, I.I., Barma, N.C.D., Hakim, A., He, Z., Fan, Z., Novoselovic, D., Maghraby, M., Gad, K.I.M., Galal, E.G., Hagras, A., Mohamed, M.M., Morad, A.F.A., Kumar, U., Singh, G.P., Naik, R., Kalappanavar, I.K., Biradar, S., Sai Prasad, S.V., Chatrath, R., Sharma, I., Panchabhai, K., Sohu, V.S., Mavi, G.S., Mishra, V.K., Balasubramaniam, A., Jalal-Kamali, M.R., Khodarahmi, M., Dastfal, M., Tabib-Ghaffari, S.M., Jafarby, J., Nikzad, A.R., Moghaddam, H.A., Ghojogh, H., Mehraban, A., Solís-Moya, E., Camacho-Casas, M.A., Figueroa-López, P., Ireta-Moreno, J., Alvarado-Padilla, J.I., Borbón-Gracia, A., Torres, A., Quiche, Y.N., Upadhyay, S.R., Pandey, D., Imtiaz, M., Rehman, M.U., Hussain, M., Hussain, M., Ud-Din, R., Qamar, M., Sohail, M., Mujahid, M.Y., Ahmad, G., Khan, A.J., Sial, M.A., Mustatea, P., von Well, E., Ncala, M., de Groot, S., Hussein, A.H.A., Tahir, I.S.A., Idris, A.A.M., Elamein, H.M.M., Manes, Y., Joshi, A.K., 2017. Strategic crossing of biomass and harvest index—source and sink—achieves genetic gains in wheat. Euphytica 213, 257. https://doi.org/10.1007/s10681-017-2040-z.
Ritz, C., Pipper, C.B., Streibig, J.C., 2013. Analysis of germination data from agricultural experiments. Eur. J. Agron. 45, 1–6. https://doi.org/10.1016/j.eja.2012.10.003.
Schaarschmidt, F., Djira, G.D., 2016. Simultaneous confidence intervals for ratios of fixed effect parameters in linear mixed models. Commun. Stat. - Simul. Comput. 45, 1704–1717. https://doi.org/10.1080/03610918.2013.849741.
Smart, R.E., Bingham, G.E., 1974. Rapid estimates of relative water content. Plant Physiol. 53, 258–260. https://doi.org/10.1104/PP.53.2.258.
Stroup, W.W., 2015. Rethinking the Analysis of Non-Normal Data in Plant and Soil Science. Agron. J. 107, 811. https://doi.org/10.2134/agronj2013.0342.
van der Vaart, A.W., 1998. Delta method. Asymptotic Statistics. Cambridge University Press, Cambridge, pp. 25–34. https://doi.org/10.1017/CBO9780511802256.
Vickers, A.J., 2001. The use of percentage change from baseline as an outcome in a controlled trial is statistically inefficient: a simulation study. BMC Med. Res. Methodol. 1.